Quiz 2-B

First and Last Name: ____________________

You have 15 minutes to complete the quiz. Calculators, notes, and other resources are **not** allowed.

1. (1 point) Fill in the last entry in the following matrix product. No work is necessary.

$$\begin{bmatrix} 3 & 7 \\ -2 & 5 \\ 2 & 9 \end{bmatrix} \begin{bmatrix} 4 & 7 \\ -1 & -3 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -13 & -29 \\ -1 & \boxed{} \end{bmatrix}$$

Solution: We do the dot product between $\begin{bmatrix} 2 & 9 \end{bmatrix}$ and $\begin{bmatrix} 7 \\ -3 \end{bmatrix}$.Then we have

$$2 \cdot 7 + 9 \cdot (-3) = -13.$$

2. (2 points) Reduce the following matrix to reduced row echelon form. Show all work, including your row operations.

Hint: You can reduce this matrix to reduced row echelon form using only two elementary row operations.

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 3 & 1 & -3 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 3 & 1 & -3 \end{bmatrix} \xrightarrow{r_3 - 3r_1 \to r_3} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{bmatrix} \xrightarrow{r_3 - r_2 \to r_3} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

QUESTION 3 IS ON THE BACK!!

3. Suppose B is the matrix where

$$\left(2B + \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 1 & -3 \end{bmatrix}\right)^T = \begin{bmatrix} 4 & 0 & 2 \\ 1 & 5 & 4 \end{bmatrix}.$$

(a) (1 point) What's the size of B? (i.e. if B if $m \times n$, find m and n).

Solution: To ensure the addition operator that is valid, B should be 3×2.

(b) (2 points) Find B. Show all work.

Solution: Use the property of matrix transpose, we have

$$2B + \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 1 & -3 \end{bmatrix} = \left(\begin{bmatrix} 4 & 0 & 2 \\ 1 & 5 & 4 \end{bmatrix}\right)^T = \begin{bmatrix} 4 & 1 \\ 0 & 5 \\ 2 & 4 \end{bmatrix}.$$

Thus,

$$2B = \begin{bmatrix} 4 & 1 \\ 0 & 5 \\ 2 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 1 \\ 1 & 0 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 5 \\ 1 & 7 \end{bmatrix},$$

and then

$$B = \begin{bmatrix} 1/2 & 0 \\ -1/2 & 5/2 \\ 1/2 & 7/2 \end{bmatrix}.$$